

"There are many products available in the market like drone products, wireless products, flex and force sensors, 3D printers and development boards, among others," says Dileep Jain, sales head, Rajguru Electronics. The most popular here are boards such as Raspberry Pi iterations, which are more than enough proof that actively engaging the students in project development from within the college itself is the way to go. The Pi is not the only hot seller, though.

Compella says, "We use Arduino boards to help students develop interest in robotics." Arduino has come up as a major solution for beginners in terms of technical education. Kishore supports the argument with, "It is more suited to hobbyists and high school students." Once they know the boards, students can go for designing more complex projects with more specialised equipment.

While college labs tend to stock these boards, students may do better with many other boards available in the market. Today, there are many online retailers that have opened up shops to sell these. "Once we show the students the many possibilities with an ARM mbed board and programmable system on chip (PSoc) Bluetooth Low Energy (BLE) chip, their interest is exponentially increased," adds Dr Joshi.

Development boards are good in the beginning where the students feel excited about clicking a few switches to see a light glow. Kishore feels that, "At some point they would have to do the programming for their own projects, hence such boards cannot be used indefinitely." Once the students get comfortable with the tech, they have to move to other solutions. "Students often move to PIC MCU to implement their projects," says Brijesh C.A., chief executive officer, Just Robotics.

Software tools continue to thrive

Ranging from simple software for beginners to complex integrated

What smart learning is

Today we see the world being smart at everything. Learning is another extension to the smart wave. Often the use of real components is not possible in classes due to one or the other reason. In such cases, adopting smart learning methods helps make learning fun and easier. Smart learning includes, but is not limited to, smart classes, use of digital content, online learning courses and such.

A smart class often uses white boards with easy access to projectors and digital study content, which help in explaining the hard-to-understand concepts to students. According to Manish Joshi, deputy chief executive officer at Sciencetech, "Some difficult subjects can be made easier with the help of smart learning." For example, an animated video of the working of a human heart can be easily understood by students compared to the traditional 2D depiction on a chart paper.

development environments (IDEs) capable of designing entire systems, we have a lot of technology available for students working on developing their skills. While schools have smart classes with digital study content to make learning easier, graduate and postgraduate colleges are also taking up smart teaching techniques using automation and simulation tools to make students understand or see the impact of certain design decisions.

This has helped in getting the attention of students in class and increased student participation in class discussions. A report by Inderpreet Kaur Chachra on the effectiveness of smart learning found a significant improvement in student performance after exposure to smart learning solutions. You can read about it on the next page in the box titled 'Case studies on smart classes.'

"Today, we have a lot of open source tools being used," says Brijesh. This increases availability of good and efficient tools for students. Today, we have various tools like EasyEDA and MeowCAD that help in electronics designing, whereas CAD tools like Solidworks and OnShape help in bringing the product to life as envisioned by the engineer.

In engineering, tools like MATLAB are very popular and already part of the syllabus in many universities. Other popular tools that students use today include National Instrument's LABVIEW, Proteus and Allegro PCB Designer.

The T-shaped engineer

There is a lot of interest in this concept of the T-shaped engineer, who has deep expertise in one domain like controls but also general knowledge of other domains. This concept helps teams of engineers to be collaborative and solve those types of problems that are inherently multi-domain in nature.

That said, free alternatives to popular paid software are gaining steam with tools like SageMath, which is an open source mathematical software. Other popular tools include CadSoft EAGLE for PCB design, FreePCB, gEDA, Quite Universal Circuit Simulator, Ngspice, Icarus Verilog and KiCad.

One area that has seen a rapid increase in interest in getting started with software development tools is in the Internet of Things (IoT), which we cover next.

Covering the IoT for beginners

"There is a growing demand for connected devices," says Compella. "Since the demand for IoT products is now higher, working with IoT development boards helps in speeding up product development."

Dr Joshi adds, "ARM mbed boards have also come as a solution for college graduates in understanding IoT products."

Such solutions that can develop products for the up-and-coming technologies and not for soon-to-be-outdated ones are ideally suited to learning. "People are spending